

FUERSTENZELL, GermanyWMO: **108950**Lat: **48.5451N**Lon: **13.3532E**Elev: **480**StdP: **95.69**Time Zone: **1.00 (EUC)**Period: **98-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-9.5	-14.5	1.1	-10.4	-11.8	1.4	-8.0	13.4	7.4	11.3	5.7	1.8	50	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	29.1	19.3	27.3	18.8	25.7	18.1	20.5	26.6	19.6	25.4	18.8	24.3	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	14.1	22.9	17.6	13.4	21.8	16.9	12.8	20.9	61.1	26.5	58.2	25.6	55.4	24.4	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.2	6.8	-14.9	31.9	2.7	1.4	-16.9	32.9	-18.5	33.7	-20.1	34.5	-22.0	35.5		
(5)			-15.3	22.2	2.6	0.8	-17.2	22.8	-18.7	23.3	-20.2	23.7	-22.1	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-1.6	-0.3	4.2	9.4	13.9	16.9	18.5	18.1	13.9	9.0	3.4	-0.8
	DBStd	8.24	4.55	4.44	4.28	4.19	3.79	3.85	3.36	3.20	3.35	3.74	3.84	3.84
	HDD10.0	1507	361	288	184	59	9	1	0	0	6	63	201	335
	HDD18.3	3616	618	522	439	268	144	73	42	43	136	288	449	593
	CDD10.0	1058	0	0	3	42	131	207	262	253	124	33	3	0
	CDD18.3	123	0	0	0	1	7	29	45	37	4	1	0	0
	CDH23.3	994	0	0	0	6	67	236	367	291	26	1	0	0
	CDH26.7	214	0	0	0	1	6	51	93	63	1	0	0	0
Wind	WSAvg	2.8	2.8	3.2	3.7	3.0	2.9	2.6	2.6	2.4	2.5	2.5	2.6	2.7
Precipitation	PrecAvg	951	75	66	73	51	89	100	110	98	75	67	71	77
	PrecMax	1320	173	146	202	103	152	153	189	264	146	169	146	168
	PrecMin	664	9	13	14	3	14	35	39	23	29	17	0	20
	PrecStd	137	43	36	51	25	40	35	41	50	34	38	37	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	11.3	17.2	23.9	27.5	30.6	31.7	31.2	25.9	21.6	15.5	9.9
		MCWB	8.6	7.4	10.0	13.7	17.8	19.6	20.0	20.3	18.3	15.6	11.7	8.0
	2%	DB	8.4	9.2	15.0	21.3	25.3	28.1	29.1	28.3	23.7	18.1	12.1	7.2
		MCWB	6.9	6.4	9.4	12.7	16.3	19.2	19.3	19.2	17.2	14.1	9.8	6.0
	5%	DB	6.3	7.4	12.9	19.2	23.2	26.2	27.2	26.5	21.9	16.2	10.2	5.6
		MCWB	5.1	5.1	8.3	11.8	15.5	18.5	18.9	18.6	16.4	13.3	8.8	4.7
	10%	DB	4.5	5.6	10.7	17.0	21.0	24.2	25.3	24.7	20.0	14.4	8.4	4.1
		MCWB	3.6	3.9	7.5	10.8	14.4	17.4	18.3	17.9	15.6	12.3	7.5	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	8.1	11.1	15.0	19.2	21.6	21.6	21.5	19.7	16.6	11.9	8.3
		MCDB	11.1	10.3	15.2	21.7	25.6	27.7	28.5	28.3	24.2	19.9	14.9	9.6
	2%	WB	7.0	6.7	9.8	13.4	17.4	20.2	20.4	20.2	17.9	14.9	10.2	6.2
		MCDB	8.3	8.9	13.7	19.4	23.2	26.1	26.7	25.9	22.2	17.1	11.6	7.0
	5%	WB	5.1	5.2	8.9	12.3	16.2	19.1	19.5	19.2	17.0	13.7	9.0	4.7
		MCDB	6.2	7.0	12.2	18.1	21.7	24.7	25.5	24.9	20.7	15.8	9.9	5.4
	10%	WB	3.6	4.0	7.7	11.1	15.1	18.0	18.7	18.4	16.0	12.6	7.5	3.5
		MCDB	4.5	5.5	10.4	16.2	19.9	22.8	24.0	23.4	19.2	14.2	8.3	4.0
Mean Daily Temperature Range	5% DB	MDBR	4.3	5.3	7.3	9.1	9.3	9.0	9.3	9.1	8.2	6.8	4.4	4.0
		MCDBR	6.0	7.8	11.5	12.6	12.9	12.5	12.9	12.5	11.6	9.5	7.2	6.0
		MCWBR	4.8	5.5	6.8	6.2	6.1	5.2	5.2	5.2	5.9	5.9	5.1	4.9
	5% WB	MCDBR	5.6	7.3	10.2	11.7	11.5	11.3	11.3	11.4	10.0	8.4	6.7	5.7
		MCWBR	4.6	5.4	6.4	6.2	5.7	5.0	5.0	5.2	5.6	5.7	5.0	4.8
Clear-Sky Solar Irradiance	taub	0.295	0.319	0.357	0.389	0.396	0.399	0.407	0.401	0.375	0.346	0.320	0.294	
	taud	2.405	2.360	2.319	2.269	2.284	2.308	2.283	2.320	2.364	2.432	2.462	2.449	
	Ebn at Noon	774	833	853	858	865	863	849	838	824	791	738	730	
	Edh at Noon	66	86	106	124	127	125	126	117	101	80	63	56	
All-Sky Solar Radiation	RadAvg	1.00	1.93	2.99	4.33	5.03	5.55	5.36	4.69	3.35	2.09	1.06	0.76	
	RadStd	0.17	0.26	0.37	0.55	0.60	0.48	0.50	0.48	0.44	0.23	0.12	0.09	